



Bachelor Thesis

The Impact of Venture Capital Funding on AI Startup Performance: A Data Analytics Study (2015-2026)

Cross-sectional study of AI startups backed by VC funding.

Supervisor: Prof. Dr. Adalbert Wilhelm

Submitted by:

Date: 31/08/2026

Saleem, Mashal

Matriculation Number: 30006850

Abstract

Artificial Intelligence has attracted increasing levels of venture capital investment, particularly following the rapid growth of generative AI. While the overall increase in AI investment has been widely documented, less is known about how VC funding relates to the performance and progression of individual AI startups.

This study examines the relationship between venture capital funding and AI startup outcomes between 2015 and 2026 by using a dataset of 705 venture-backed AI startups obtained from Crunchbase Pro. The analysis combines descriptive statistics, data visualization, OLS regression, one-way ANOVA, an independent-samples t-test and regional subgroup regression. Startup performance is measured using employee count as a proxy. The study also examines changes in funding over time, the relationship between funding and employee count, and differences in funding across startup progression stages.

The findings show a substantial increase in AI startup funding activity, particularly from 2023 onwards. Overall, the findings suggest that VC funding is closely related to the growth and progression of AI startups. However, the results should be interpreted as relationships rather than evidence of causation. The study contributes to the understanding of how VC funding is associated with measurable startup outcomes during a period of rapid expansion in the AI sector.

Statutory Declaration

Family Name, Given/First Name	Saleem, Mashal
Matriculation number	30006850
What kind of thesis are you submitting: Bachelor-, Master- or PhD-Thesis	Bachelor Thesis

English: Declaration of Authorship

I hereby declare that the thesis submitted was created and written solely by myself without any external support. Any sources, direct or indirect, are marked as such. I am aware of the fact that the contents of the thesis in digital form may be revised with regard to usage of unauthorized aid as well as whether the whole or parts of it may be identified as plagiarism. I do agree my work to be entered into a database for it to be compared with existing sources, where it will remain in order to enable further comparisons with future theses. This does not grant any rights of reproduction and usage, however.

This document was neither presented to any other examination board nor has it been published.

German: Erklärung der Autorenschaft (Urheberschaft)

Ich erkläre hiermit, dass die vorliegende Arbeit ohne fremde Hilfe ausschließlich von mir erstellt und geschrieben worden ist. Jedwede verwendeten Quellen, direkter oder indirekter Art, sind als solche kenntlich gemacht worden. Mir ist die Tatsache bewusst, dass der Inhalt der Thesis in digitaler Form geprüft werden kann im Hinblick darauf, ob es sich ganz oder in Teilen um ein Plagiat handelt. Ich bin damit einverstanden, dass meine Arbeit in einer Datenbank eingegeben werden kann, um mit bereits bestehenden Quellen verglichen zu werden und dort auch verbleibt, um mit zukünftigen Arbeiten verglichen werden zu können. Dies berechtigt jedoch nicht zur Verwendung oder Vervielfältigung.

Diese Arbeit wurde noch keiner anderen Prüfungsbehörde vorgelegt noch wurde sie bisher veröffentlicht.

Bremen, 31.08.2026

Place, Date

Mashal Saleem

Signature

Declaration of Artificial Intelligence (AI) Usage

I hereby declare that artificial intelligence (AI) tools were used responsibly in the preparation of this thesis. All AI-generated content has been critically reviewed, verified, and edited by me to ensure its accuracy, academic integrity, and alignment with the research objectives.

- **AI tools used:** ChatGPT & Claude.ai
- **Purpose of use:** Language refinement, idea structuring and data visualization
- **Extent of use:** Helped debug non-essential code components.

No AI system was used to generate or falsify data, conduct analysis, or produce content presented as original research.

I take full responsibility for the accuracy and integrity of all content in this thesis.

Bremen, 31.08.2026

Place, Date

Mashal Saleem

Signature

Contents

Abstract	i
1. Introduction	1
1.1 Motivation	1
1.2 Problem Statement	2
1.3 Process Summary	3
2. Literature Review	4
2.1 AI Landscape	4
2.2 Venture Capital Investment in AI	4
2.3 Venture Capital, Startup Growth and Progression	5
2.4 Regional Differences in Venture Capital	6
2.5 Research Gap	6
3. Data and Methodology	7
3.1 Data Sources	7
3.2 Dataset Description	7
3.2.1 Dataset Construction	7
3.2.2 Final Sample	8
3.3 Data Cleaning and Preprocessing	9
3.3.1 Data Range Filtering	9
3.3.2 Duplicate Removal	9
3.3.3 Missing Critical Data Removal	9
3.3.4 Venture Capital Filtering	10
3.3.5 Geographic Classification and Filtering	10
3.3.6 Feature Engineering	10
3.3.7 Completeness Check	10
3.4 Tools and Technologies Used	11
3.5 Data Analysis Techniques	11
3.5.1 Descriptive and Trend Analysis	11
3.5.2 Multiple Linear Regression	12
3.5.3 One-Way ANOVA and Independent-Samples t-test	12
3.5.4 Regional Subgroup Regression	12
4. Results, Analysis and Discussion	13
4.1 RQ1	13
4.2 RQ2	18
4.3 RQ3	20
4.4 RQ 4	22

5. Challenges, Limitations and Future Work.....	23
5.1 Challenges Encountered.....	23
5.2 Limitations	24
5.3 Future Research Directions	25
6. Conclusion.....	26
7. References	27

1. Introduction

1.1 Motivation

Venture capital investment in artificial intelligence has grown rapidly in recent years, reaching levels that would have been difficult to imagine only a few years ago. According to OECD, AI firms accounted for 61% of global VC investment in 2025. This shows a great increase from 2022 when AI accounted for around 30% of global VC investment. [1]

Although the recent growth of AI may appear sudden, the technology itself is the result of decades of development, from early rule-based systems to machine learning, deep learning and more recently large language models built on transformer architectures. However, the public release of ChatGPT in November 2022 became an important turning point bringing generative AI into mainstream public and business attention and contributing to a new wave of interest and investment in the sector. [2]

This growth raises an important question about the role of venture capital in the development of AI startups. While larger amount of funding may provide startups with the resources to hire employees, develop technology and expand their operations, raising more funding does not necessarily guarantee better performance or long-term success. Startups that are already growing quickly may also be likely to attract additional investment. Therefore, it is important to better understand the relationship between venture capital funding and measurable startup outcomes.

Most existing discussions of the AI investment boom focus on the overall growth of the sector, such as total investment volumes, major funding rounds, or differences between countries and regions. Less attention has been given to examining these relationships at the level of individual AI startups. This study is motivated by the need to move beyond the overall funding trend and examine how differences in venture capital funding relate to startup size, progression, and outcomes.

1.2 Problem Statement

Despite the rapid increase in VC funding in AI startups, there is still limited evidence on how differences in funding relate to the outcomes of individual AI companies. Existing reports provide useful information about the overall growth of AI investment, but investment figures alone do not explain whether startups that raise more capital also demonstrate greater organizational growth or progress further through their development.

This creates several related questions. First, it is unclear how the funding of AI startups has evolved across different periods. Second, while the venture capital is expected to provide resources that can support startup growth, the relationship between the amount of funding raised and measurable performance is not straightforward. Higher funding may be associated with larger employee counts, but the strength and shape of this relationship require empirical examination. Third, differences in funding may be related to how far a startup has progressed, including whether it remains in an early stage, reaches later funding stages, or achieves an exit through an acquisition or IPO. Finally, these relationships may differ across regions because startups operate within different investment environments and venture capital ecosystems. The main problem addressed in this study is therefore the lack of startup-level analysis examining the relationship between venture capital and progression of AI startups.

To address this problem, the study investigates four research questions:

- **RQ1:** How has venture capital funding in AI startups evolved between 2015 and 2026?
- **RQ2:** What is the relationship between the amount of venture capital funding a startup has raised and its current performance? Does this relationship increase steadily, or level off at higher funding amounts?
- **RQ3:** Does the amount and stage of venture capital funding affect whether an AI startup survives, gets acquired, goes public, or shuts down?
- **RQ4:** Does the relationship between venture capital funding and AI startup performance differ across regions (e.g., North America, Europe, Asia-Pacific)?

1.3 Process Summary

The study followed a structured process consisting of data collection, data cleaning, variable preparation, exploratory analysis, statistical analysis, and interpretation of findings.

The original dataset was constructed using Crunchbase Pro and contained information on 1,000 AI startups. The companies were selected using defined inclusion criteria related to AI industry classification, venture capital funding, and the study period. The data included variables related to funding, funding rounds, employee size, founding year, funding status, IPO status, and company location. The datasets were then cleaned and prepared for analysis. This process included removing observations that did not meet the study criteria and addressing missing or incorrectly formatted values.

The final cleaned dataset contained 705 AI startups, and all analyses in the study were conducted using this sample. For the analysis, several variables were constructed from the original Crunchbase data. Employee count was used as a proxy for startup performance and employee-size ranges were converted into midpoint values to allow quantitative analysis. Startups were grouped into founding cohorts for the analysis of funding over time. For the analysis of startup progression, a continuity status variable was created to classify companies as Early Stage, Late Stage or Successful Exit.

Different analytical techniques were then applied to answer the four research questions. RQ1 used descriptive analysis and visualization to examine changes in total funding. RQ2 used OLS regression to examine relationship between total funding and employee count. RQ3 used a one-way ANOVA to compare log-transformed funding across the three continuity groups and RQ4 repeated the regression analysis separately across different regions to examine relationship between funding and employee count.

2. Literature Review

2.1 AI Landscape

Artificial Intelligence has developed from a relatively specialized area of research into a major technical and commercial sector. While AI has existed for decades, recent activities in generative AI and machine learning have significantly increased their practical applications across industries. At the same time, the rapid growth of AI has captured significant investor attention. The commercial potential of generative AI, combined with its possible application across a wide range of industries has made the sector attractive to both venture capital and larger technology companies. SBC Asset Management highlights this broader transformation of AI from an emerging technology into an increasingly important area of commercial and investment activity. [3]

The release of ChatGPT in late 2022 represented an important turning point in this development. Although the technologies behind generative AI had been developing for several years, ChatGPT made these capabilities widely accessible to the public. Such a breakthrough demonstrated how AI could be used in everyday and commercial contexts. This contributed to increased attention from businesses, policy makers, and investors. [2]

2.2 Venture Capital Investment in AI

Venture capital has played an increasingly important role in the growth and commercialization of AI. Developing AI technologies, particularly advanced models and infrastructure, often requires massive investment before a company can generate revenue. Venture capital can therefore provide startups with the resources needed to hire specialized employees, develop products, access computing capacity and scale their operations. Investment in AI has increased considerably over the past decade. Earlier OECD research found that global venture capital investment in AI companies grew rapidly between 2012 and 2020, with the United States and China emerging as dominant investment markets. [1]

Recent insights from VC investors also suggest that the AI investment market has begun to mature. Following the initial excitement surrounding generative AI, interest has started to extend towards application-based AI companies. These are particularly developing specialized solutions for industries such as healthcare, finance, legal services and accounting [4]. Evidence from more specialized AI sectors also reflects the broader growth of VC investment in AI. VC funding in AI-driven biopharmaceutical startups substantially increased between 2010 and 2024 [5]. This provides evidence that the expansion of AI investment was extended into specialized industries.

2.3 Venture Capital, Startup Growth and Progression

Venture capital is commonly viewed as an important source of financial support for high-growing startups. In addition to providing financial resources, VC can support companies as they expand their operations and progress through different stages of development. However, the relationship between VC and startup outcomes is not straightforward. Startups may grow after receiving funding, but investors may also be more likely to fund companies that already demonstrate strong growth potential.

The issue has been examined in previous research on the relationship between VC financing and startup growth and found that startup employment growth increased both before and after venture capital financing events. [6] This relationship is particularly relevant when employee count is used as an indicator of startup performance. For private companies, detailed financial information such as revenue and profitability is often unavailable. Employee numbers can therefore provide a practical measure of organizational scale.

VC financing is also characterized by its staged nature. Rather than receiving all capital at once, startups may progress through several rounds of financing with investors re-assessing company's prospects at different stages. One of the study describes this process as one in which the continuation of financing depends on the practicality and eventual exit outcomes. [7]

2.4 Regional Differences in Venture Capital

Pantea and Tkacik's systematic review of 34 firm-level studies across 17 European countries suggests that venture capital generally has a positive impact on business growth and investment. The review however shows that the impact of venture capital differs depending on the country and the type of investor involved. This suggests that venture capital does not operate in the same way everywhere and that its success is influenced by the wider investment and business environment [8]. A study showed clear differences in VC funding between Europe and United States. European startups in the space sector typically raised less funding, completed fewer funding rounds and took longer to secure investments than US startups. They were also less likely to achieve successful exits through an IPO or acquisition. [9]

2.5 Research Gap

While there have been many studies on general startup population and VC investment, very few research has been made specifically on AI startups. This is particularly because AI landscape is still relatively new space. At the same time, fewer studies that focus on AI investments, often examined broader investment trends or specific transitions between funding stages. Therefore, this study addressed these gaps by examining VC funding at the individual startup level within a sample of AI companies. It investigated how funding has evolved, examined relationship between total funding and employee count, compared funding levels across stages of startup progression and tested whether the funding-employee relationship varies across different regions.

3. Data and Methodology

3.1 Data Sources

This study relied on data from Crunchbase, a widely used commercial database that provided information on AI startups, private companies, funding rounds, investors, and other firm-level characteristics. Crunchbase was chosen as the sole data source to ensure consistency throughout the analysis. The decision to rely on a single database was made to ensure consistency across the dataset and to avoid potential differences in definitions and classifications that may arise when combining data from multiple providers. Data collection was conducted using a paid Crunchbase Pro subscription. The platform's Advanced Search function was used to identify companies based on the selection criteria of the study, after which the relevant company-level data was exported for further analysis. This approach allowed the data to be collected systematically and, in a format, suitable for the cross-sectional design of the study.

3.2 Dataset Description

3.2.1 Dataset Construction

The initial exported dataset consisted of 1000 startups and 21 columns consisting of key metrics. Since the AI market is fairly new in terms of huge VC investment (the space has only picked up pace in the last few years especially after the rise of generative AI tools like ChatGPT in late 2022), the research infrastructure i.e public datasets built specifically for studying it hasn't really caught up yet. Another reason is a lot of firms haven't made it to IPO yet, so a lot of metrics remain private. Hence, the dataset was built from scratch.

To do this, data was collected directly from Crunchbase, using its Advanced Search export tool. A paid Crunchbase Pro subscription was used to access this data rather than a free trial. This was an important decision as the free version of Crunchbase offers very limited data and doesn't allow you to export as a file. This helped ensure that data collected was of a higher and more

reliable quality than what would have been available through the free tier. Because Crunchbase shows live, constantly updated company data rather than a fixed historical record, the initial export of 1000 companies (21 columns) reflected a real-time snapshot of the platform as it stood in July 2026.

3.2.2 Final Sample

The dataset comprised company-level records for artificial intelligence (AI startups) founded between 2015 and 2026. Companies were identified using Crunchbase's Industry Groups classification, especially firms tagged under 'Artificial Intelligence (AI)' and related categories, combined with 'Funding Type', filtered restricting the sample to only VC-backed firms ('Seed', 'Early-Stage Venture', 'Late-Stage Venture', 'IPO' and 'M&A' funding statuses). The final analytical sample consisted of 705 startups. Each record includes the following core attributes exported directly from Crunchbase:

Variable	Description
Organization Name	Company identifier
Founded Date	Year of company founded
Headquarters Region	Geography location descriptor
Total Funding Amount (USD)	Cumulative funding raised to date
Number of Funding Rounds	Total count of completed funding
Last Funding Date/Amount/Type	Details of most recent funding round
Operating Status	Active, Acquired, or Closed Status
Funding Status	Seed, Early Stage, Late Stage, IPO, M&A
Number of Employees	Categorical employee count range
IPO Status	Public or Private listing status

To support the study's four research questions, several variables were derived from raw fields during preprocessing, including founding-year cohorts, broad geographic region groupings, a numeric employee-count estimate, a log-transformed funding measure, and a simplified continuity/progression status. These derived variables are described in detail in section 3.3.

The final sample is distributed across three funding cohorts (2015-2018, n=144; 2019-2022, n=249; 2023-2026, n=134; Asia-Pacific, n=76).

3.3 Data Cleaning and Preprocessing

Data cleaning was conducted using Python (pandas' library) following a structure, sequential pipeline, summarized below. The original export contained 1,000 company records; each stage of cleaning reduced this to the final analytical sample of 705 startups.

3.3.1 Data Range Filtering

Records were filtered to preserve companies founded only between 2015 and 2026, removing entries with irrelevant founding rates outside the study period (*1,000 → 842 records*).

3.3.2 Duplicate Removal

Exact duplicate company entries were identified and removed based on organization name (*842 → 839 records*).

3.3.3 Missing Critical Data Removal

Records missing essential funding information (Total Funding Amount or Number of Funding Rounds) were excluded, as these variables are central to all four research questions (*839 → 826 records*).

3.3.4 Venture Capital Filtering

The sample was restricted to companies with a Funding Status of Seed, Early-Stage Venture, Late-Stage Venture, IPO & M&A, excluding Private Equity-backed firms and companies with unclassified funding status. IPO & M&A companies were kept, as these represent startups that were venture-backed prior to a successful exit (*826 → 720 records*).

3.3.5 Geographic Classification and Filtering

A broad regional variable (*hq_region*) was constructed by matching keywords with the Headquarters Regions field to one of three categories: North Africa, Europe, or Asia-Pacific. Records with unclassifiable or extremely low-frequency regional data (“Rest of World”) were excluded to support meaningful regional comparison in RQ4 (*720 → 715 records*).

3.3.6 Feature Engineering

Categorical employee ranges (e.g. “51-100”) were converted to numeric midpoint estimates for use in regression analysis. A natural log-transformation of Total Funding Amount was created to address the substantial right-skew in the funding distribution. This was done to test for non-linear relationships. Funding status was recorded into three broader categories – Early Stage, Late Stage, and Successful Exit (combining IPO and M&A) to serve as the primary outcome variable for RQ3. This substitution was necessary because Operating Status exhibited minimal variation in the dataset (over 95% of records classified as ‘Active’), a pattern attributed to Crunchbase’s tendency to under-update the status of inactive or discounted companies.

3.3.7 Completeness Check

Records with missing values across any of the key variables required for analysis were removed. A full cleaned dataset of 705 records was retained for analysis, rather than drawing a smaller sub-sample, to preserve maximum statistical power across all for research questions, particularly for the regional sub-group comparisons in RQ4. Missing data in secondary variables such as (Estimated Revenue Range, Patents Granted, Monthly Visits Growth, Total Products Active) were retained as missing rather than imputed, consistent with standard practice for data that is missing due to non-disclosure rather than data entry error. These fields were excluded from primary regression analyses. (*715 → 705 records*).

3.4 Tools and Technologies Used

All data cleaning and analysis was conducted using Python 3 within a Jupyter Notebook environment (VS Code). The following libraries were used:

- **Pandas and NumPy** for data loading, cleaning, transformation and numerical computation.
- **Statsmodels** for regression modelling (Ordinary Least Squares).
- **SciPy** for statistical hypothesis testing (ANOVA, independent-samples t-test).
- **Matplotlib** and **Seaborn** for data visualization.

Raw data was exported from Crunchbase's Advanced Search interface in CSV format and processed entirely within this Python environment. The complete, commented analysis notebook is provided in Appendix ().

3.5 Data Analysis Techniques

Several methods such as Descriptive Analysis, Multiple Linear Regression, One-Way ANOVA and Independent-Samples t-test, Regional Subgroup Regression were used in this study.

3.5.1 Descriptive and Trend Analysis

For RQ1, startups were grouped into two different ways: (1) by founding cohort (2015-2018, 2019-2022, 2023-2026), to compare cumulative funding and number of rounds across startup generations, and (2) by the calendar year of their most recent funding round, to capture actual year-by-year deal-making activity. For each grouping, the mean, median, and count of relevant variables (Total Funding Amount, Number of Funding Rounds) were calculated using pandas' *groupby()* and *agg()* functions. A cross-tabulation *pd.crosstab()* was used to calculate the percentage distribution of funding-stages (Funding Status) within each founding cohort.

3.5.2 Multiple Linear Regression

Ordinary Least Squares (OLS) regression was used, implemented via the *statsmodels* Python library. The dependent variable was *employees_midpoint* (a numeric estimate of company size, derived from Crunchbase's categorical employee-count ranges). The independent variable was Total Funding Amount, tested in two forms: (1) its raw dollar value, and (2) a natural-log-transformed version (*log_funding*). Founding year was included as a control variable in both models, since older companies have simply had more time to grow their teams, independent of funding received. The two models' R^2 values were compared to assess whether the funding-employee relationship is closer to linear or exhibits diminishing returns.

3.5.3 One-Way ANOVA and Independent-Samples t-test

A one-way ANOVA via compared average log-transformed funding across the three continuity status categories (Early Stage, Late Stage, Successful Exit) to test whether funding amount differs meaningfully depending on how far a startup has progressed. A supplementary independent-samples-t-test (through *scipy.stats.ttest_ind()*) compared average log-transformed funding between publicly listed (IPO) and privately held companies specifically using the IPO status field.

3.5.4 Regional Subgroup Regression

The RQ2 regression model i.e funding predicting employee_count, controlling for founding year was run three times. Once for each regional subgroup (North America, Europe, Asia-Pacific) using the same model specification each time. The resulting coefficients, R^2 values and p-values were compared across the three regional models.

4. Results, Analysis and Discussion

4.1 RQ1

How has venture capital funding in AI startups evolved between 2015 and 2026?

RQ1 established the broader funding landscape within which the rest of the study operates. Before testing relationships between funding and performance (RQ2-RQ4), it was necessary to first understand whether, and how, funding activity itself has changed over the study period. Three things were examined: (1) how much funding different generations of startups have raised in total, and how many rounds it took them; (2) how much money was actually deployed into AI startups in each individual calendar year; and (3) how the mix of funding stages (Seed through IPO/M&A) shifts depending on when a startup was founded.

Key findings:

- Startups founded earlier i.e from 2015 to 2018 show higher average cumulative funding ($\approx$ 920M) and more completed rounds (7.09 on average) than those founded most recently i.e from 2023 to 2026 ($\approx$ 427M, 2.99 rounds. This reflects the additional time older startups have had to raise successive rounds, not necessarily a stronger funding environment during their founding period.
- Actual year-by-year deal activity shows a clear acceleration: from a single \$15.2M deal in 2019, activity grew steadily through the early 2020s, then accelerated sharply from 2023 onward, reaching 428 deals totalling roughly \$173.0B by 2026.
- The proportion of startups at later funding stages including Late-Stage Venture, IPO, and M&A is highest in the oldest cohort and lowest in the most recently founded cohort. This pattern is likely to reflect, at least in part, the different levels of maturity across the cohorts. Startups founded earlier have had more time to progress through successive funding rounds

and potentially reach later-stage financing, acquisition or an IPO, whereas companies in the newest cohort are generally still at an earlier stage of their development.

The bar chart below shows for each calendar year (2019-2026) with the height of each bar representing how many startups in the sample has their most recent funding round in that year. The bars are short and roughly flat through 2019-2022, then grow taller each year from 2023 onward, with the tallest bar in 2026. This visually represents the sharp increase in deal-making activity identified in RQ1.

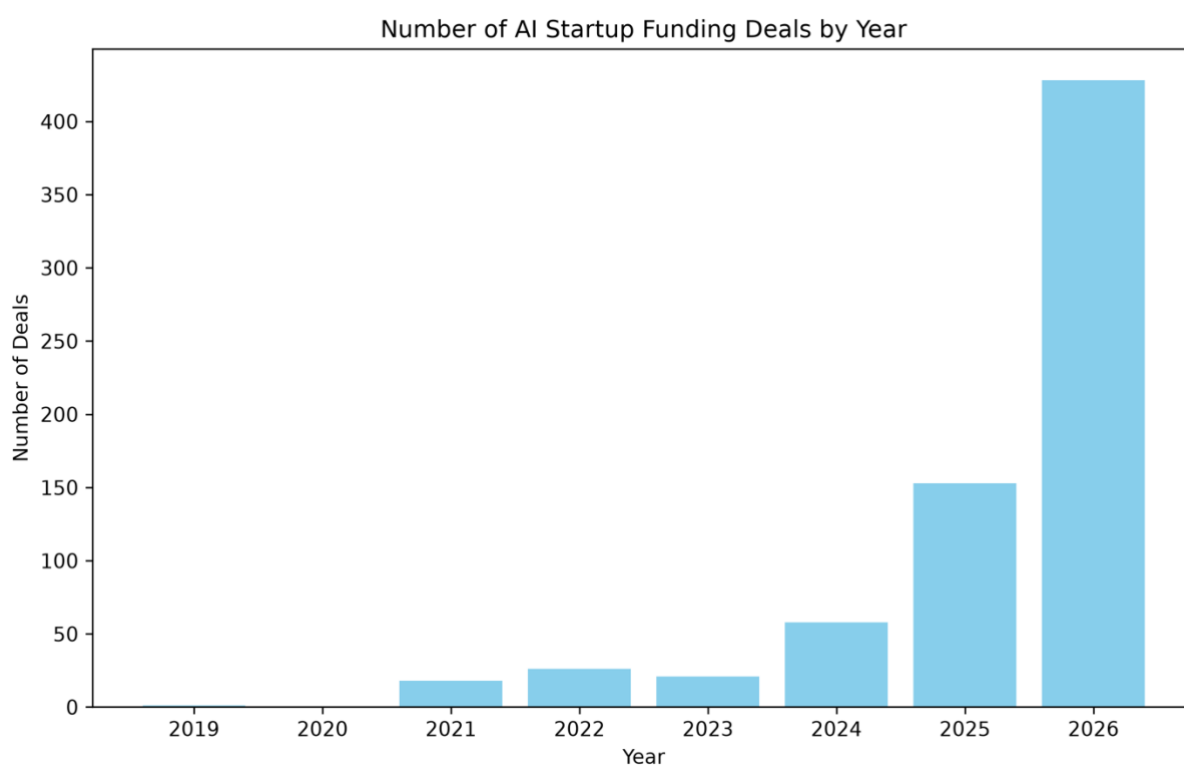


Fig 1. Number of AI Startup Funding Deals by Year

The line graph below shows total amount of VC funding (in billions of USD) deployed into AI startups plotted year by year. The line nearly stays flat and close to \$0B from 2019 through 2023. It starts climbing modestly in 2024 (~6B). Then jumps sharply to ~20B in 2025. It then rises dramatically to ~173B in 2026. The figure tells the same “boom” story as Fig 1 (deal

count) but from the dollar value angle instead of the number of deals. Altogether, the two charts reinforce each other.

Fig 1 (deal count): more AI startups are closing funding rounds each year, especially since 2023

Fig 2 (total funding): the total amount of USD being deployed into these rounds is growing even faster, not just more deals but bigger deals too.

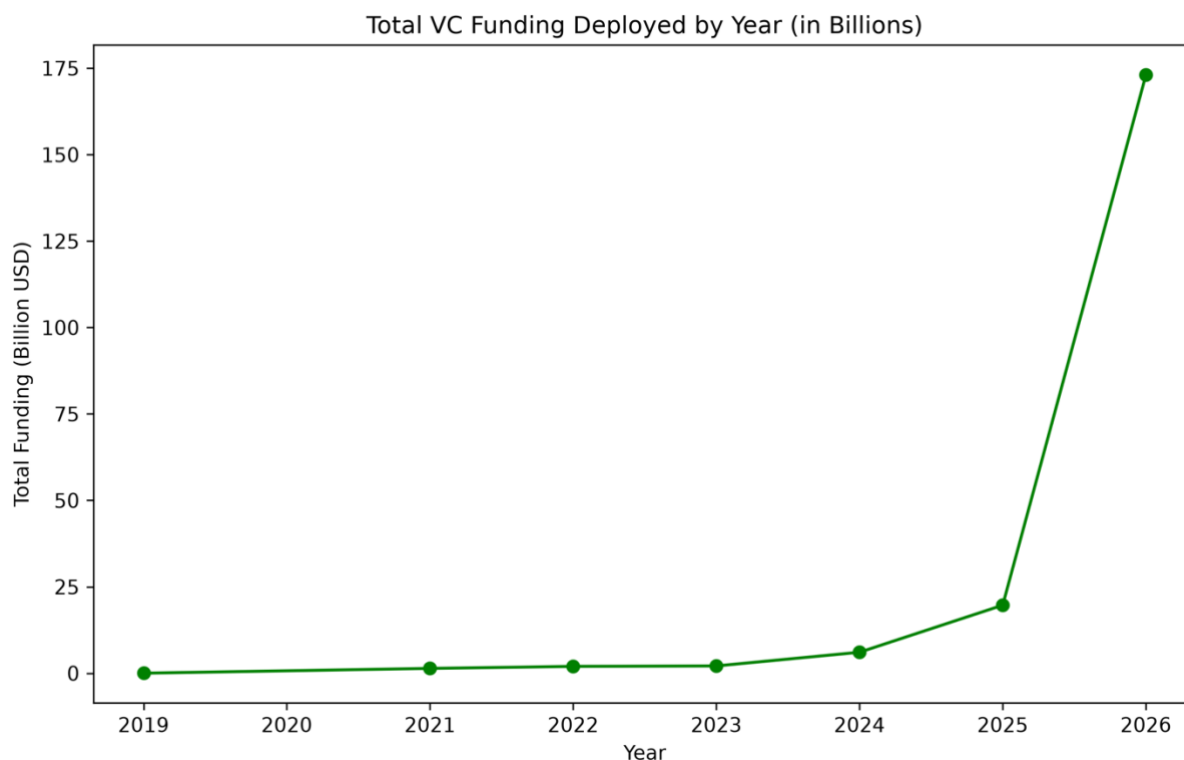


Fig 2. The total amount of VC funding (in billions of USD)

A bar chart of each of the three funding cohorts (2015-2018, 2019-2022, 2023-2026) with height represents the average cumulative funding raised by startups in that cohort.

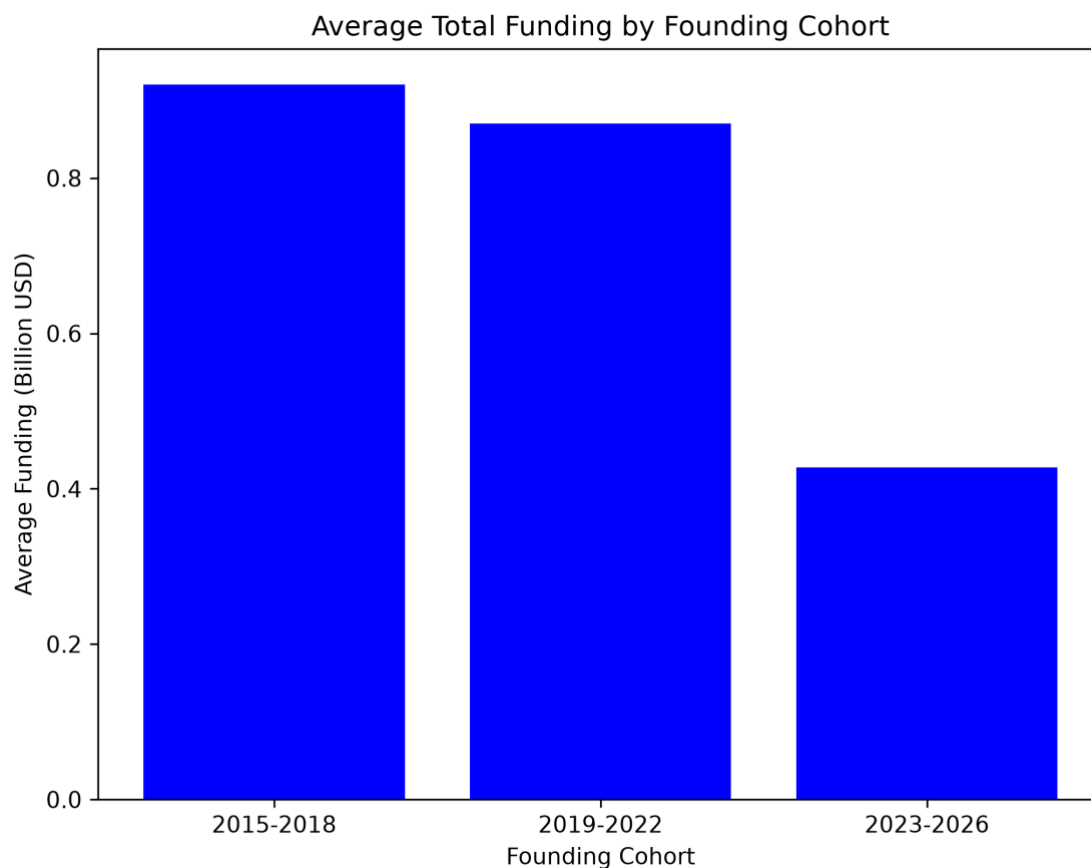


Fig 3. Average total funding by funding cohort

Fig 4 below is a stacked bar chart. Each bar represents one funding cohort (2015-2018, 2019-2022, 2023-2026) and each colored segment shows what percentage of startups in that cohort currently sit at a given funding stage. Late-Stage Venture (green) dominates (roughly 50%) in the **2015-2018** cohort. IPO (orange) makes up around 10%, M&A (red) presents around 8%, and Early-Stage Venture presents around 32%.

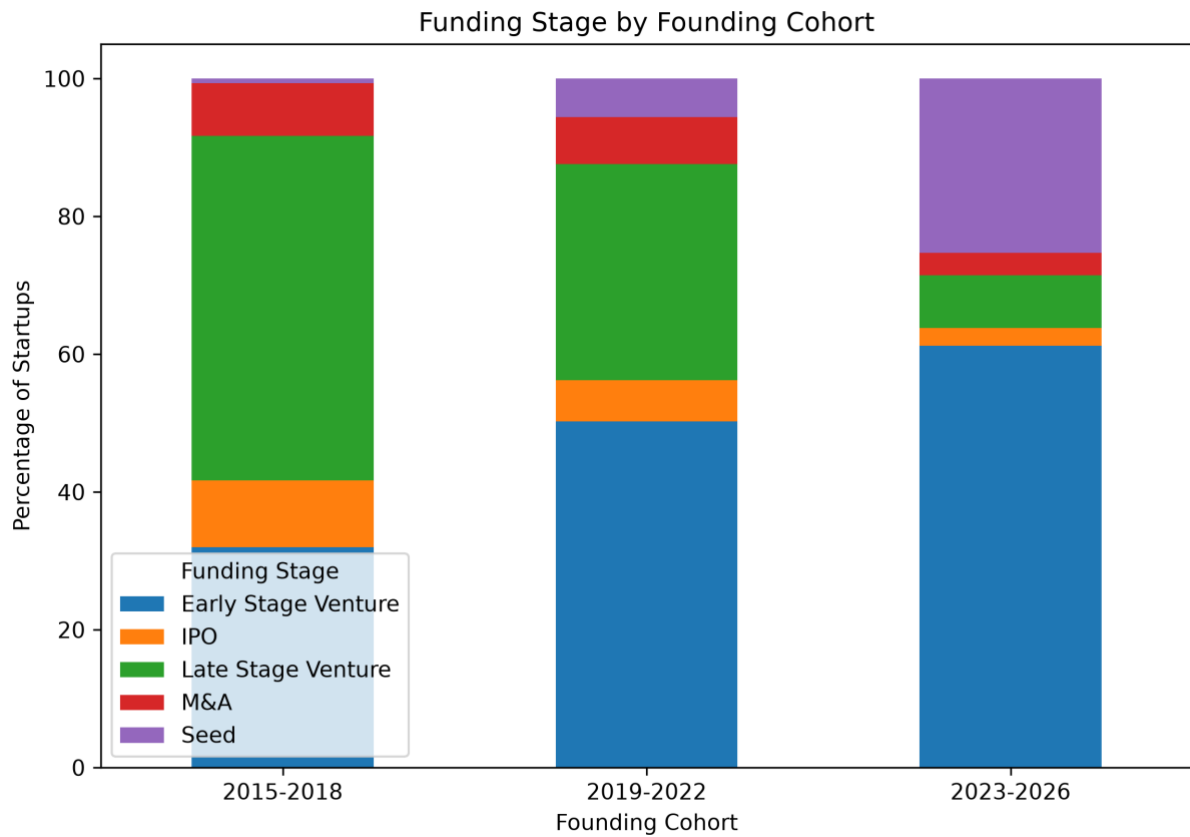


Fig 4. The funding stage distribution by each cohort.

For the cohort **2019-2022**, blue becomes the largest segment (~50%), green substantially shrinks (~30%), IPO and M&A both remain smaller than the previous cohort. For the newest cohort **2023-2026**, blue extremely dominates (~61%), purple (Seed) appears meaningfully for the first time (~25%), while all others remain minimal.

4.2 RQ2

What is the relationship between the amount of venture capital funding a startup has raised and its current performance? does this relationship increase steadily, or level off at higher funding amounts?

Two OLS regression models were estimated to examine the relationship between venture capital funding and employee count. The first model used the original “Total Funding Amount”, while the second used a log-transformed funding variable. Both models controlled for the year in which the startup was founded and included 705 observations. As shown in Table 3.4, the raw model provided a better overall fit to the data, explaining 37.1% of the variation in employee count ($R^2=0.371$), compared with 26.6% for the log-transformed model ($R^2=0.266$). The adjusted R^2 values showed a similar pattern. In both models, the funding variable was statistically significant ($p<0.001$), indicating a positive relationship between the amount of funding raised and employee count.

Measure	Raw Funding Model	Log-transformed funding model
No. of observations	705	705
R-squared	0.371	0.266
Adjusted R-squared	0.369	0.264
F-statistic	207.0	127.5
Funding coefficient	6.34×10^{-8}	163.99
Funding p-value	< 0.001	< 0.001
Founded year coefficient	-72.30	-50.29
Founded year p-value	< 0.001	< 0.001

Table 3.4

The founded year variable was negative and statistically significant in both models ($p < 0.001$). This suggests that, after accounting for funding, startups founded more recently tended to have lower employee counts than older startups in the sample.

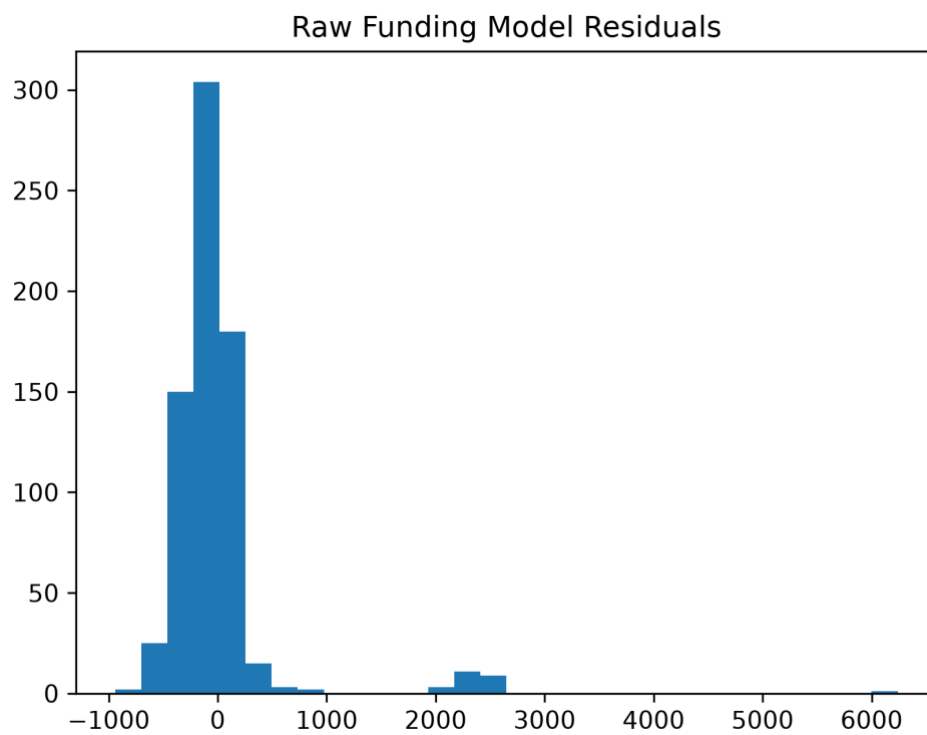


Fig 5. Residual Plot: Raw Funding Model.

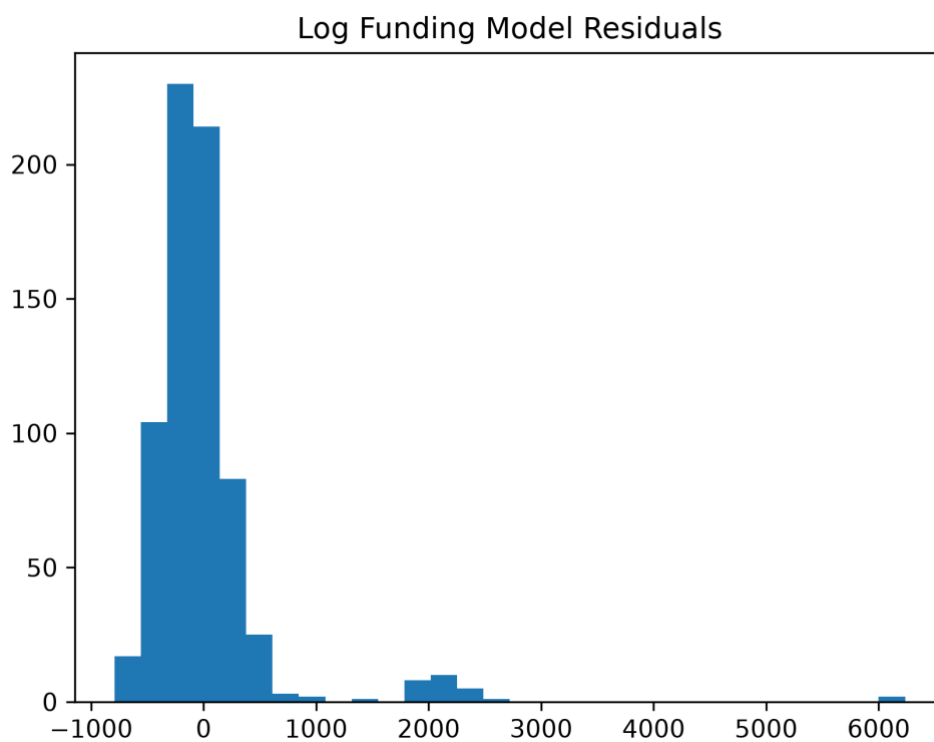


Fig 6. Residual Plot: Log-Transformed Funding Model.

Figures 5 and 6 show the distribution of residuals for the raw funding and log-transformed funding models. In both cases, most residuals are concentrated within a relatively narrow range, although several observations have much larger positive residuals. This suggests that there are some startups whose employee counts are substantially higher than predicted by the model.

The distributions are also clearly skewed rather than normally distributed. Applying the log transformation to funding does not change this pattern, as large positive residuals remain in both models.

4.3 RQ3

Does the amount and stage of venture capital funding affect whether an AI startup survives, gets acquired, goes public, or shuts down?

For this question, a variable `continuity_status` was constructed classifying each startup as Early Stage, Late Stage, or Successful Exit (IPO or M&A) based on its current Funding Status. Average funding or round counts were compared across these three groups using a one-way ANOVA. A supplementary t-test separately compared funding levels between companies that had and had not completed an IPO.

Average funding differs significantly across the three continuity groups ($F = 106.52$, $p < .001$): Early-Stage startups have raised far less on average ($\approx \$184\text{M}$) than Late Stage ($\approx \1.49B) or Successful Exit ($\approx \$1.87\text{B}$) startups. Companies that completed an IPO raised significantly more funding on average ($\approx \$2.34\text{B}$) than privately held companies ($\approx \$590\text{M}$) ($t = 4.70$, $p < .001$).

The Fig 7. below shows distribution of log-transformed funding across the three continuity stages. The box plot shows that Early-Stage startups generally have lower funding levels than the other two groups. In comparison, both Late-Stage and Successful Exit startups have higher

median funding levels, although their distribution overlap. The successful exit group also shows a wider spread, suggesting greater variation in the amount of funding raised by startups that reached an IPO or M&A outcome.

These patterns are consistent with the ANOVA results, which found a statistically significant difference in average log-transformed funding across all the three continuity groups ($F = 106.52$, $p < .001$). Early-Stage raised less funding on average (approx. \$184 million) than Late-Stage (approx. \$1.49B) and successful exit (approx. \$1.87B). Overall, the results suggest that higher funding levels are associated with progression beyond the early stage.

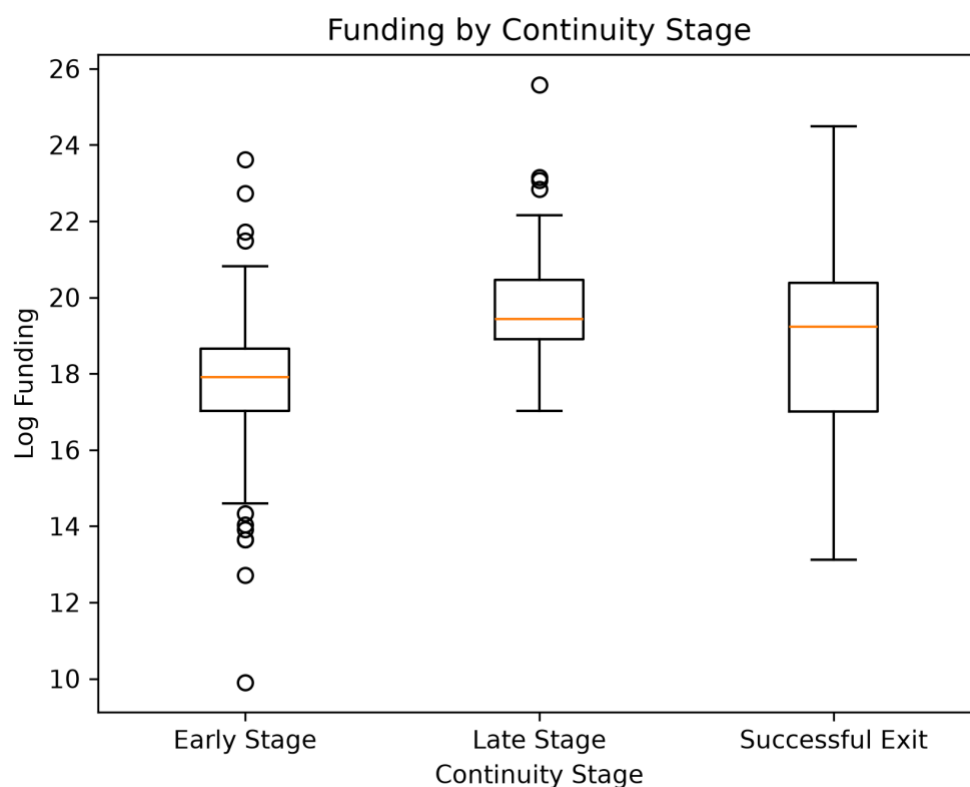


Fig 7. Funding by Continuity Stages (Early-Stage, Late-Stage, Successful Exit)

4.4 RQ 4

Does the relationship between venture capital funding and AI startup performance differ across regions?

The RQ2 regression model was re-run separately for each of the three regional subgroups in the sample. The relationship was statistically significant in every region (all $p < .01$). North America showed the strongest relationship (coefficient = 183.91), followed by Europe (139.16), with Asia-Pacific showing the weakest, though still clearly significant (122.07). Model fit (R^2) was similar across all three regions (0.265-0.313).

Fig 8 below shows the relationship between log-transformed funding and employee count across three regional groups (North America, Europe, and Asia Pacific). Each point represents a startup, while the fitted regression line shows the overall relationship between funding and employee count within each region. Across all three regions, the regression line shows a positive upward trend. This indicates that startups with higher levels of funding generally tend to have larger employee counts. The positive relationship was statistically significant in North America, Europe and Asia-Pacific (all $p < .01$). The relationship appears strongest in North America, which has the largest regression coefficient ($\beta = 183.91$), followed by Europe ($\beta = 139.16$). Asia-Pacific had the smallest coefficient ($\beta = 122.07$). This suggests that, within the sample, an increase in funding was associated with a larger increase in employee count in North America than in other two regions.

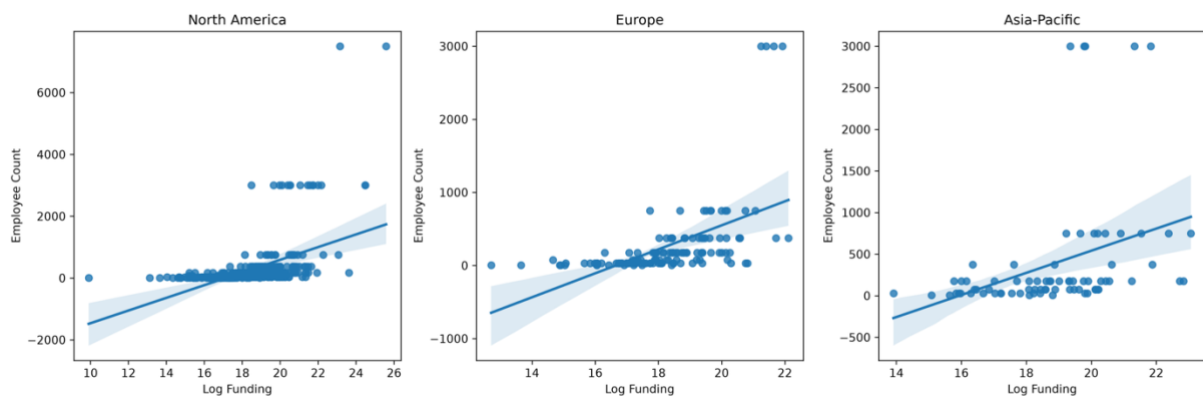


Fig 8. Relationship Between Funding and Employee Count Across Regions

5. Challenges, Limitations and Future Work

5.1 Challenges Encountered

The study encountered several practical and methodological challenges. Many of these were related to the nature of the AI startup sector itself, which is developing rapidly and does not yet have the same level of established research infrastructure or publicly available data as more mature industries.

One of the main challenges was absence of a ready-made dataset that combined information on AI startups, their funding and relevant performance measures. As a result, the dataset had to be constructed specifically for this study using Crunchbase's Advanced Search and export tools. This required decisions about which companies should be included, how AI startups should be defined, which funding types were relevant, and which time-period should be covered.

Data collection was also affected by the export limits of the Crunchbase Pro subscription. Because the number of exports was limited, the search criteria had to be planned carefully before exporting the data. This made the data collection more restrictive, as it was not possible to repeatedly export larger samples and refine the selection afterwards.

Some data cleaning challenges were caused by formatting issues in the exported data. For instance, certain employee ranges such as "11-50" and "1-10" appeared as date-like values, including "Nov-50" and "10-Jan". These values were manually identified and corrected before the employee ranges could be converted into numerical midpoint estimates for the regression analysis.

5.2 Limitations

The main limitation of this study is that the data represent a single snapshot of startups rather than periodic observations of the same startups over time. Although the study covers companies founded between 2015 and 2026 and examines funding activity across different years, the underlying company-level data were collected at one point in time from Crunchbase. In other words, the dataset does not track changes in individual startup's employee counts, performance or other outcomes at regular intervals. Therefore, the study could not construct a true longitudinal or time-series analysis of how individual startups developed over time. This is particularly important when interpreting the regression results. While higher funding was found to be related to higher employee counts, the cross-sectional design does not show whether funding led to employee growth. This finding should therefore be interpreted as relationship rather than causal effect.

Employee count was used as a proxy for startup performance due to absence of consistent financial information such as revenue and profitability. This was due to many privately held companies in the dataset. While the employee count provides an indication of organizational growth and size, it does not capture all aspects of startup performance. A company with more employees is not necessarily more profitable or more successful.

Lastly, dataset contains several companies with very high funding amounts and employee counts. These observations are not data errors but such observations can have a strong influence on regression results and contribute to the highly skewed distribution as seen in the results.

5.3 Future Research Directions

The most important direction for the future research would be to use periodic or time-series data that follows the same startup over time. For example, future studies could collect employee counts, revenue or other performance measures at regular intervals and compare these changes with the timing of funding events. This would make it possible to examine whether startup performance changes after a company receives funding, rather than simply comparing companies at different stages at one point in time. Future research could also examine individual countries rather than broad regional groups. This would allow researchers to explore whether differences in venture capital markets, government support, regulation or startup ecosystems influence the relationship between funding and startup performance.

6. Conclusion

The study examined the impact of venture capital funding on the performance and progression of AI startups between 2015 and 2026. Using a final sample of 705 venture-backed AI startups obtained from Crunchbase Pro, the study addressed four research questions covering changes in AI funding over time, the relationship between funding and startup performance, differences in funding across stages of progression, and regional differences in the funding-performance relationship. The findings show that venture capital activity in AI startups increased from 2023 onwards. Both the number of funding deals and total amount of funding increased sharply in the more recent years. At the same time, startups founded in earlier cohorts had, on average, raised more cumulative funding and completed more funding rounds than newer startups. This appears to reflect the fact that older companies have had more time to progress through successive funding rounds, rather than necessarily indicating that the funding environment was stringer when they were founded.

The analysis also founded a positive and statistically significant relationship between the amount of venture capital funding raised and employee count, which was used as a proxy for startup performance. The raw funding model provided a better fit to the data than the log-transformed model, explaining 37.1% of the variation in employee count compared with 26.6%. For startup progression, funding levels differed significantly across Early-Stage, Late-Stage and Successful Exit companies. Early-Stage startups had substantially lower average funding than both Late-Stage and Successful Exit startups. Companies that completed an IPO also showed significantly higher average funding than privately held companies. Overall, these findings suggest that higher funding levels are closely related to progression beyond the early stages of startup development. However, the study shows that VC funding is strongly related to several measurable outcomes among AI startups. The study is based on observational, cross-sectional data and therefore identifies relationships rather than causal effects.

7. References

- [1] (2026). *Venture capital investments in artificial intelligence through 2025*. OECD.
https://www.oecd.org/en/publications/venture-capital-investments-in-artificial-intelligence-through-2025_a13752f5-en/full-report.html
- [2] Adamec, V. (2025). *A Brief History of AI and the Rise of ChatGPT*. gauss algorithmic.
<https://www.gaussalgo.com/knowledge-base/a-brief-history-of-ai-and-the-rise-of-chatgpt>
- [3] (2026). *Venture Capital: Investing in the Age of AI*. HSBC Asset Management.
<https://no.assetmanagement.hsbc.com/en/institutional-and-professional-investor/news-and-insights/investing-in-the-age-of-ai>
- [4] (2026). *The AI Investigating Landscape: Insights from Venture Capital*. Federal Reserve Bank of San Francisco.
<https://www.frbsf.org/research-and-insights/blog/sf-fed-blog/2026/02/11/the-ai-investing-landscape-insights-from-venture-capital/>
- [5] Bazaz, A., Ji, Y., Socal, M.P., & Kang, S-Y. (2026). *Trends in Venture Capital Investment in AI-Driven Biopharmaceutical Startups*. *Journal of Medical Internet Research*.
<https://pubmed.ncbi.nlm.nih.gov/41740146/>
- [6] Davila, A., Foster, G., & Gupta, M. (2023). *Venture capital financing and the growth of startup firms*. *Journal of Business Venturing*. 689-708.
https://www.sciencedirect.com/science/article/abs/pii/S0883902602001271?utm_source=chatgpt.com

[7] Greenwood, J., Han, P., & Sanchez, J.M. (2018). *Financing Ventures*. *National Bureau of Economic Research*. 24808.

https://www.nber.org/papers/w24808?utm_source=chatgpt.com

[8] Pantea, S. (2022). *Venture capital and high-tech start-ups in Europe: a systematic review of the empirical evidence*. *Venture Capital*. 435-458.

https://www.tandfonline.com/doi/full/10.1080/13691066.2024.2315069?utm_source=chatgpt.com

[9] Colombo, M.G., Lucini-Paioni, A., & Tofan, B. (2026). *Mapping venture capital investments in the (new) space economy with large language models: a comparative analysis of Europe and the United States*. *Journal of Industrial and Business Economics*.

<https://link.springer.com/article/10.1007/s40812-026-00401-7>